The Meaning of the Volcano: Gravitational and Circumferential Energy

To explain the volcano's meaning—how it relates to the laws of physics—I have to get a bit technical.

For the sake of simplicity, let's pretend the *Endurance* is moving in Gargantua's equatorial plane. (For the *Endurance*'s nonequatorial trajectory the ideas are the same but because the black hole is not spherical, the details are more complicated.) The volcano analogy neatly encapsulates the true physics of the critical orbit and Gargantua's trajectory. To explain how, I need two physics concepts: the *Endurance*'s *angular momentum,* and its *energy*.

After tidal forces pry it apart from Mann's planet, the *Endurance* has a certain amount of angular momentum (its circumferential speed around Gargantua times its distance from Gargantua). The relativistic laws tell us that this angular momentum remains fixed (conserved) along the *Endurance*'s trajectory; see Chapter 10. This means that, as the *Endurance* plunges toward Gargantua, with its distance from Gargantua decreasing, its circumferential speed increases. This is similar to an ice-skater, whose whirling speed increases when she moves her arms in (Figure 27.4).